

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1857

Austin Dam 138 kV
April 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1857. The Interconnection Customer proposes a 314.0 MW wind facility interconnecting at the Austin Dam 138 kV substation in the Pennsylvania area. The request was received on August 13, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 531 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,030,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Wolverine Crk / L L Wilkins Substation 138 kV
Contingency: Westwood Generation Llc / Coronet 500 kV
Loading: 126.5% (Rating: 197 MVA)
DFAX: 6.4%
Pre-project loading: 116.3%
2. Facility: Berlin 5 / Austin Dam 138 kV
Contingency: Ft Gage / Austin Dam 138 kV
Loading: 115.7% (Rating: 252 MVA)
DFAX: 100.0%
Pre-project loading: 8.9%
3. Facility: Wolverine Creek / Yorkshire 138 kV
Contingency: Coronet / Westwood Generation Llc 500 kV
Loading: 105.6% (Rating: 219 MVA)
DFAX: 6.4%
Pre-project loading: 96.4%
4. Facility: Ft Age / Austin Dam 138 kV
Contingency: Coronet / Westwood Generation Llc 500 kV
Loading: 104.5% (Rating: 384 MVA)
DFAX: 78.5%
Pre-project loading: 40.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wolverine Crk / L L Wilkins Substation 138 kV	Rebuild line with higher-capacity conductor	\$2,356,000	100% (\$2,356,000)
Austin Dam / Berlin 5 138 kV	Rebuild line with higher-capacity conductor	\$5,175,000	100% (\$5,175,000)
Yorkshire Substation / Wolverine Crk 138 kV L	Reconductor line	\$1,978,000	100% (\$1,978,000)
Ft Age / Austin Dam 138 kV	Reconductor line	\$521,000	100% (\$521,000)
	Total allocated to project		\$10,030,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.